

Deepak Mallampati

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PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) specializing in LLM-powered agent systems, evaluation frameworks, and production agentic workflows. Master's degree (Kent State University). Proven track record building multi-agent pipelines with schema-validated contracts, RAG systems with >90% retrieval-grounded alignment, and evaluation pipelines that drove ~35% precision improvement and >30% hallucination reduction. Experienced shipping AI features to production (FastAPI/Docker) with measurable quality metrics. Passionate about agent reasoning, orchestration, and building evaluation frameworks that reliably hill-climb toward better helpfulness and reliability.

CORE COMPETENCIES

LLM Agent Development • Multi-Agent Orchestration • Evaluation Frameworks & Benchmarks • Retrieval-Augmented Generation (RAG) • Agentic Workflow Design • LangChain / LlamaIndex • Prompt Engineering & Structured Outputs • Python / FastAPI / Flask • PyTorch / Hugging Face Transformers • Docker / MLOps / LLMOps • Experimentation Pipelines • Production AI Safety & Guardrails

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS Healthcare | USA

- Built **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Designed **multi-agent pipeline architecture** with schema-validated JSON contracts between agents to prevent cascading hallucinations; implemented structured reasoning traces and audit-ready outputs.
- Built **Retrieval-Augmented Generation (RAG) evaluation framework**; implemented recursive semantic chunking, transformer-based embeddings, and automated retrieval quality measurement; improved contextual retrieval precision by ~35%, reduced irrelevant retrieval by >30%, and achieved >90% retrieval-grounded response alignment in evaluation.
- Developed **LLM evaluation pipelines** measuring hallucination rates, grounding alignment, and response stability; drove >30% **hallucination reduction** through iterative agent behavior analysis and prompt refinement.
- Packaged LLM/ML inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by ~30%.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics, Applied NLP | Hyderabad, India

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; reduced manual review time by **60-70%** (hours to <5 minutes per video).
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85% in controlled tests.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP | Hyderabad, India

- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors >30% and improved release cycle time ~35-40%.
- Designed performance dashboards using **SQL Server DMVs and Grafana**; cut incident recurrence ~25% and improved root-cause resolution speed ~30%.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Python, LangChain, RAG, vector search, FastAPI, Docker, multi-agent orchestration

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript app with coordinated multimodal API orchestration. Introduced **intermediate structured prompt transformation layers** - improved contextual alignment ~**30%** and first-pass success rate ~**25-30%** over naive direct-prompt orchestration. Demonstrates agent reasoning and output quality improvement through structured intermediary steps.

Python, FastAPI, React, TypeScript, multi-provider LLM orchestration, structured outputs

Agentic Pixel Character Synthesis & Animation Engine

LLM-based agent orchestrator that decomposes high-level prompts into structured generation tasks, coordinating Stable Diffusion fine-tuning (LoRA), ControlNet pose conditioning, and latent-space consistency layers. Demonstrates multi-step agent task decomposition and skill routing.

PyTorch, Hugging Face Diffusers, FastAPI, Docker, LLM orchestration, ControlNet, OpenPose

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped

Full-stack browser extension integrating **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider payload handling, seven-day translation cache, per-domain selector configs for 29 sites. Demonstrates production AI integration with multi-model orchestration and reliability engineering.

JavaScript, Manifest V3, multi-provider AI APIs, Chrome Web Store

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Agent & LLM Systems: LangChain, LlamaIndex, RAG, agentic workflows, multi-agent orchestration, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search

Languages & Frameworks: Python, FastAPI, Flask, React, TypeScript, SQL (T-SQL, PostgreSQL, SQLite)

ML Libraries: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost, Diffusers, OpenCV, YOLOv8

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision